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Laser-Synthesized Bismuth Nanoparticles with Controlled Efficiency of Photothermal Conversion for Multimodal Theranostics

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Abstract—The dependence of the efficiency of photothermal conversion of laser-synthesized bismuth (Bi) nanoparticles (NP) on size characteristics for applications in photothermal therapy is studied. It was shown that a decrease in the mode and half-width of the size distribution causes an increase in the photothermal conversion efficiency. Despite a decrease in the extinction in the relative transparency window of biological tissues (650–950 nm), Bi NPs of smaller sizes exhibit a more pronounced photothermal response caused by an increased contribution of absorption to the extinction. The unique combination of physicochemical properties of laser-synthesized Bi NPs forms a new promising alternative to conventional sensitizers of photothermal therapy of oncological diseases.

Keywords: bismuth, nanoparticles, laser ablation in liquids, photothermia

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1. INTRODUCTION

The use unique properties of nanomaterials offers prospects for the development of classical applications and the formation of fundamentally new applications in both diagnostics [1] and therapy of oncological diseases [2]. Of particular interest for biomedical applications are sensitizers of multimodal theranostics [3].

The record absorbance of X-ray radiation (7.38 cm^2/g at 90 keV), pronounced metallic properties, and a relatively low toxicity of nanoparticles (NP) based on bismuth (Bi) open new attractive prospects for the simultaneous development of radiation and photothermal theranostics methods [4, 5].

Conventional methods of Bi NP synthesis are based on chemical synthesis approaches using toxic precursors and complex solutions for synthesis initiation [6]. In this context, pulsed laser ablation in liquid (PLAL) which allows obtaining ultrapure NPS with controllable physicochemical properties is a promising attractive for classical methods [7]. The versatility, simple performance, and high efficiency of the PLAL cause its wide acceptance in nanophotonics, power engineering, and biomedicine [8–10].

In this work, we study the dependence of the photothermal conversion efficiency in the region of relative transparency of biological tissues of laser-synthesized Bi NPs on their size characteristics. The direct energy conversion of light radiation to thermal energy depends on the optical absorption efficiency of the influence sensitizer. In this case, the absorption is caused by both fundamental parameters of a material and size, morphological, and structural properties of NPs themselves. It was shown demonstrated that a decrease in the mean and size distribution half-width of laser-synthesized Bi NPs from 120 ± 89 to 18 ± 9 nm leads to an increase in the photothermal conversion efficiency from 30.4% to 38.8%, respectively, which can be caused by the dominant contribution of absorption to the extinction for small-size NPs.

2. MATERIALS AND METHODS

Bi NPs were synthesized by the femtosecond PLAL method. To this end, the crystalline Bi target (GoodFellow, USA, 99.999% purity) was fixed vertically in a glass cell filled with 15 ml of acetone, at a distance of 3 mm from the cell wall. As a radiation source, a Yb:KGW TETA-10 fs laser (1025 nm, 250 fs, 25 μJ, 100 kHz, Avesta, Russia) was used. The synthesis procedure duration was 30 min.

Size and morphological properties of Bi NPs were characterized using a MAIA 3 (Tescan, the Czech Republic) scanning electron microscope (SEM) at an accelerating voltage of 30 kV. Optical extinction spectra were measured in the spectral range of 350–1100 nm using a ML122 (SOL Instruments, Belarus) spectrophotometer.

Bi NP size fractions were separated by centrifuging. To estimate photothermal properties of Bi NPs, they were irradiated in a glass cell with an optical pathlength of 10 mm by continuous laser radiation (830 nm, 940 mW, the beam diameter is 2 mm). The temperature was measured using a FLIR C3 (FLIR Systems, Wilsonville, OR, USA) thermal imaging camera. The NP concentration was measured by a gravimetric method by weighing a Bi target before and after PLAL. The NP photothermal conversion efficiency (η) was calculated by the method developed in [11]:

$$\eta = \frac{\Delta T_{\max} m_{\text{acetone}} C_{\text{acetone}} B}{(I_0 - I_c)} \times 100\%,$$

where $\Delta T_{\max}$ is the maximum temperature difference of the NP solution and an ambient medium, m_{acetone} and C_{acetone} are the mass and specific heat capacity of a liquid medium of a colloidal solution (acetone), B is the temporal constant defined as the slope of the dependence of the cooling time on logarithm of $(\Delta T_{\max}/\Delta T(t))$, I_0 is the initial radiation power, and I_c is power after passing the cell with a solution.

3. RESULTS AND DISCUSSION

NPs with the highest efficiency of exciting radiation conversion to the thermal energy in the relative transparency window of biological tissues (650–950 nm) are optimum for photothermal theranostics. To determine the dependence of the photothermal conversion efficiency on the size characteristics, various fractions of laser-synthesized Bi NPs were selected by centrifuging. The size properties were characterized by analyzing SEM images of NPs. As a result of laser ablation and size selection, colloidal solutions of spherical Bi NPs with a size distribution mode in the range of 18–120 nm and a half-width of 9–89 nm, respectively (see Fig. 1) were obtained. Then the fractions corresponding to NP sizes 18 ± 9, 33 ± 20, 65 ± 42, and 120 ± 89 nm were denoted by XS (Fig. 1a), S (Fig. 1b), M (Fig. 1c), and L (Fig. 1d), respectively.

The size characteristics of nanomaterials have an effect on their colloidal stability, the efficiency of cellular internalization and circulation duration in the blood flow, and an optimum size range for in vivo biomedical applications is 30–60 nm [12]. An increase in the diameter of metal NPs usually leads to a shifting the localized plasmon resonance to lower frequencies and broadening the resonance peak [13]. In this case, the NP extinction cross section also increases, the largest contribution to which at sizes more than 50 nm begins to be made by scattering [13].

The optical extinction spectra of Bi NPs of various size fractions are shown in Fig. 2a. Bi NP heating and cooling profiles were studied upon excitation with continuous laser radiation in the relative transparency window of biological tissues (830 nm, 1 W) (Fig. 2b). The Bi NP concentration in the experiment for all presented sizes was 200 μg/mL. An analysis of cooling profiles of colloidal (Au) solutions of Bi NPs (Fig. 2c) made it possible to determine the efficiency of the exciting radiation energy conversion to the thermal energy for each size fraction at a wavelength of 830 nm (Fig. 2d) according to [11].

One of the key parameters determining the applicability of nanomaterials for sensitization of photothermal theranostics, are their optical properties. A decrease in the mean and half-width of the size distribution (350–1100 nm), in particular in the relative transparency window of biological tissues. However, as the NP diameter exceeds 50 nm, scattering begins to make a dominant contribution to extinction, which has a negative effect on the efficiency photothermal conversion [13].

Laser-synthesized Bi NPs with smaller mean sizes and half-width size distributions showed higher photothermal conversion efficiencies reached η = 38.8% for the XS NP fraction, which exceeds a similar value for classical plasmon materials (Au NP: η = 10.2%) and nanocomposites on their basis (hybrid Fe-Au NPs: η = 38.0%) [3]. In this case, the lowest photothermal conversion efficiency (η = 30.4%) was shown by Bi NPs of fraction L having the largest mode and size distribution half-width among all size

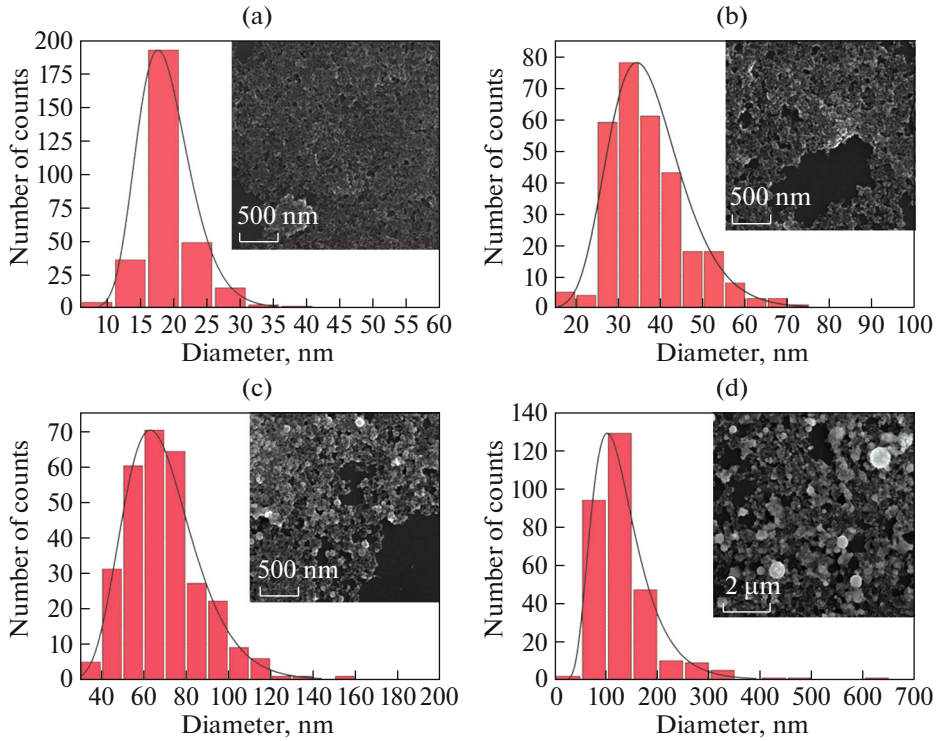

Fig. 1. SEM images and the size distribution of laser-synthesized Bi NPs of various size fractions: (a) XS, (b) S, (c) M, and (d) L.

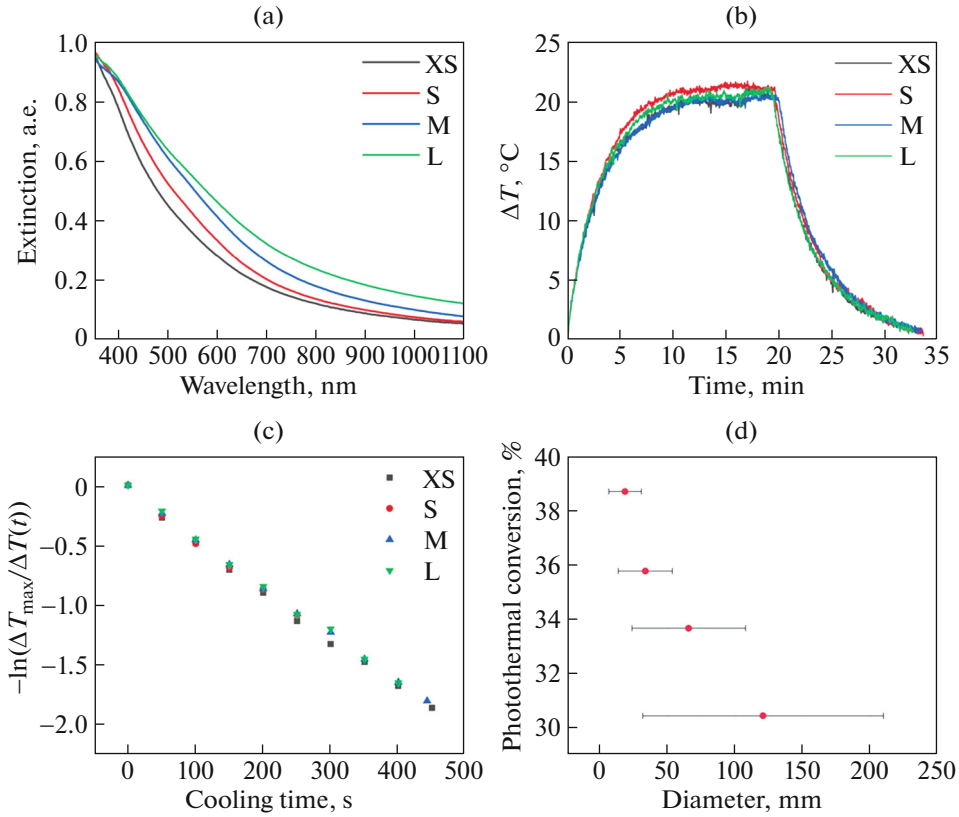

Fig. 2. (a) Extinction spectra, (b) heating curves, and (c) logarithmic cooling curves of various size fractions of laser-synthesized Bi NPs. (d) Dependence of the photothermal conversion efficiency of Bi NPs on their size characteristics (means are red points, half-widths are black curves).

fractions under consideration. Such differences can be explained by different contributions of scattering to optical extinction at the excitation wavelength.

4. CONCLUSIONS

In this study, laser synthesis, size separation, and characterization of Bi NP properties were performed.

For the first time, the controllability of the photothermal conversion efficiency with varying size characteristics of Bi NPs was studied. It was found that a decrease in the mean and size distribution half-width of laser-synthesized Bi NPs leads to an increase in the conversion efficiency of the exciting radiation energy to the thermal energy. Bi NPs with a size distribution mode of 120 nm have higher values of the extinction in the transparency window of biological tissues relative to NPs with a mode 18 nm; however, their heating profile has not significant differences, which can be caused by the different contribution of absorption to the extinction at the excitation wavelength. It was shown that Bi NPs with optimized size characteristics feature the photothermal conversion efficiency in the relative transparency window of biological tissues, exceeding similar values for classical sensitizers of photothermal theranostics ($\eta = 38.8\%$). The high absorbance of X-ray radiation in combination with pronounced and controllable photothermal effect of laser-synthesized Bi NPs form a new promising platform for developing the methods for sensitizing multimodal theranostics of oncological diseases.

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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